

2(a)			3
		description	
		name of white blood cell	
	A large cell that has a bean-shaped (kidney-shaped) nucleus. It can develop into a macrophage.	monocyte ;	
	A cell that has a large spherical nucleus and little cytoplasm. It responds to non-self antigens.	lymphocyte ; I B- / T-	
	A cell that has a lobed nucleus. It is phagocytic.	neutrophil ;	
2(b)	allows, easier / quicker, flow / movement, of blood (compared to human red blood cells) ; <i>plus any two explanations from :</i> lozenge / torpedo / elliptical / AW, shape of camel red blood cells (vs, biconcave disc of human red blood cells) ; camel red blood cells are smaller (than human red blood cells) ; ora larger number of camel red blood cells (per unit volume) ;		3
2(c)(i)	any three from: <i>in context of better adapted for uptake in lungs</i> (at high altitude) haemoglobin of llama more highly saturated with oxygen (than haemoglobin of human) ; I ref. to higher saturation in partial pressures found in tissues at <u>6.4 kPa</u> / 3500 m, oxygen saturation is, <u>86%</u> for llama and <u>79 / 80%</u> for human / 6 / 7% higher oxygen saturation for llama ; llama haemoglobin has a higher affinity for oxygen than human haemoglobin ; more oxygen, transported / delivered to tissues ; (so) sufficient oxygen unloaded in tissues to satisfy demand ;		3
2(c)(ii)	curve drawn to the right of the human dissociation curve ;		1

Question	Answer	Marks
2(c)(iii)	any three from: (metabolically active organs) release more carbon dioxide / partial pressure of carbon dioxide is higher ; ref. to decreased affinity of haemoglobin for oxygen / explained ; e.g. ref to haemoglobinic acid more oxygen released ; for more aerobic respiration / more ATP production ; <i>look for 'more' once</i>	3

Question	Answer	Marks
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